

SHUVAGATA NATH SOUMMA

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Education

Bangladesh University of Engineering and Technology (BUET)
Bachelor of Science in Mechanical Engineering
CGPA: 3.32/4.00

Dhaka, Bangladesh
Apr. 2019 – Jul. 2024

Standardized Test Scores

Graduate Record Exam (GRE)

Feb. 2026

Overall Score **319** | Quantitative Reasoning **169/170** | AWA **4.0/6.0**.

IELTS Academic

Sep. 2025

Overall Score **8/9** | Listening 8.5/9 | Reading 7.5/9 | Writing 7.5/9 | Speaking 7.5/9.

Research Interests

- Autonomous Systems
- Smart Manufacturing
- Intelligent Control
- Intelligent Robotics
- Industrial Automation
- Reinforcement Learning
- Multi-Agent Systems
- Production Optimization

Selected Projects

Hierarchical Reinforcement Learning for Multi-Robot System Coordination

Feb. 2026 – Apr. 2026

- Built a Gymnasium-compatible multi-robot warehouse simulator with PPO-based HRL scheduling, A* navigation, and congestion-aware path-overlap modeling.
- Compared HRL with greedy, congestion-aware greedy, and random baselines using reward, delivery throughput, and collision/congestion metrics.
- Improved over greedy baselines in the final experiment, achieving 5.9 average deliveries and reducing collision/congestion events to 746.1.

Autonomous Warehouse Delivery Robot in a Dynamic Environment

Mar. 2026 - Apr. 2026

- Developed an autonomous warehouse robot for shelf-to-shelf delivery using ROS 2, TurtleBot3, and Python-based mission control.
- Built a dynamic warehouse simulation in Gazebo with custom shelf aisles, moving human obstacles, LiDAR sensing, and collision avoidance.
- Implemented SLAM mapping, AMCL localization, and Nav2 path planning in RViz for autonomous navigation and delivery execution.

DRL-Based Production Scheduling Optimization for Stochastic Manufacturing Systems

Mar. 2026

- Built a stochastic smart-manufacturing simulation with machine breakdowns, demand variation, inventory constraints, and multi-stage production dependencies.
- Developed a Deep Q-Network (DQN) scheduler and explored multi-agent reinforcement learning for adaptive factory control under uncertainty.
- Benchmarked RL policies against heuristic baselines using throughput, WIP inventory, invalid actions, and demand satisfaction metrics.

IoT-Based Industrial Quality Monitoring and Defect Detection

Jan. 2025 – Mar. 2025

- Developed a sensor-based system using ESP8266 to measure real-time pressure in finished products for leakage testing and validation.
- Implemented cloud-based data acquisition and time-series analysis to identify abnormal pressure behavior and detect leakage patterns.
- Performed data-driven root cause analysis linking pressure anomalies to underlying process issues, enabling process modifications that reduced customer complaints by 30%.

Explainable Industrial Defect Detection using Computer Vision and Deep Learning

Aug. 2025

- Developed an industrial defect detection pipeline using OpenCV, PyTorch, and the MVTec AD dataset for automated classification of defective and non-defective products.
- Evaluated multiple image preprocessing techniques including CLAHE, denoising, and sharpening to analyze their impact on CNN-based defect detection performance.
- Integrated Grad-CAM explainability and generated evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrices for interpretable defect analysis.

Publications

1. **S. N. Soumma**, S. T. Ahmed, M. Ehsan, and C. R. Shaishab, “A Study of Enhancing Thermal Comfort and Development of an Intelligent Efficient Cooling System for Parked Car Interiors,” *Proceedings of the 9th BSME International Conference on Thermal Engineering (ICTE)*, 2024.
Presented at the conference.
2. U. W. Tabassum, **S. N. Soumma**, A. Sengupta, and T. Tabassumut, “Design and Fabrication of a Cost-Effective, Refreshable Mechanical Braille Display for Visually Impaired Students in Bangladesh,” *Proceedings of the 14th International Conference on Mechanical Engineering (ICME)*, 2023.
Presented at the conference.

Technical Skills

Programming: Python, MATLAB

Machine Learning: Reinforcement Learning, Deep Learning

Computer Vision: OpenCV, PyTorch

Robotics Frameworks: ROS 2, Gazebo, RViz

Navigation Systems: SLAM Mapping, AMCL Localization

Simulation Tools: Gazebo Simulation, ANSYS

CAD Software: SolidWorks, Fusion 360

Technical Tools: LaTeX, Tecplot 360

Industry Experience

Assistant Manager – Quality

Sep. 2024 – Present

A.O. Smith Corporation

- Built a Python-based pattern detection system for recurring manufacturing quality complaints to support repeat defect identification and root-cause analysis.
- Analyzed molding and material-change processes using production and quality data to recommend process corrections and reduce defects.
- Conducted data-driven quality analysis to identify defect trends, investigate process deviations, and support CAPA activities for manufacturing process improvement.

Research & Development Engineer

Jun. 2023 – Oct. 2023

Spectrum Engineering Consortium (Pvt.) Ltd.

- Designed mechanical subsystems for an autonomous EOD robot, focusing on structural performance and actuator integration.
- Performed structural and dynamic analyses (torque, vibration) using ANSYS to evaluate system response under operational loading.
- Supported CNC fabrication and prototype testing, iterating designs based on experimental validation and performance evaluation.

Leadership Experience

Team Leader, AUTOMAESTRO: Formula Student Team of BUET

Nov. 2022 – Jul. 2024

Led a 40-member multidisciplinary team in Formula Student competitions; coordinated CNC manufacturing with BITAC, supervised chassis and component fabrication, and managed international sourcing of critical subsystems.

Vice President, BUET Automobile Club

May. 2023 – Jul. 2024

Directed technical and organizational initiatives, including BUET AutoFest and engineering workshops; mentored students in CAD modeling and FEA fundamentals to develop design and analysis capabilities.

Class Representative, Dept. of Mechanical Engineering, BUET

Apr. 2019 – Apr. 2023

Represented a cohort of 180 students, facilitating communication with faculty and coordinating academic activities and issue resolution.

Event Organizer, IMechE, BUET

Apr. 2019 – Jul. 2024

Organized seminars, technical talks, and competitions, engaging national and international speakers and participants.

Achievements & Certifications

Academic Excellence Scholarship, Government of Bangladesh

2018 – 2024

Best Project Proposal, Energy & Environment Coursework, BUET

2024

Technical Speaker, IMechE Seminar, BUET

2023

4th Place (International), European Rover Challenge (ERC), Poland

2022

11th Place, Lap Time Simulation, Formula Student UK

May. 2022